



Solar power

Task A: Calculating the cost of electricity

- 1) A house with 4 or more bedrooms consumes an average of 11.8 kWh of electricity per day. In 2024, the average cost of electricity in the UK could be calculated as:

24.5 p per kWh consumed + 60 p daily standing charge

- a) How much electricity would this house consume per year?
- b) What would be the daily cost of electricity for this house?
- c) What would be the monthly cost of electricity for this house (based on 30.5 days)?
- d) What would be the annual cost of electricity for this house?

Task B: Calculating possible savings with solar panels

- 1) A house consumes an average of 11.8 kWh of electricity per day.
The cost of obtaining electricity from a supplier company can be calculated using the following formula:

cost = 24.5 p per kWh consumed + 60 p daily standing charge

The owners install 8 solar panels, each with a power rating of 380 watts.

The panels get 4.5 hours of peak sunlight per day, on average.

The performance of the panels is reduced by 25% due to factors such as shade.

How much money do the occupants save each year on their electricity bills by having solar panels?

$$E = PH$$

E = electricity produced (kWh)

P = power rating of panel (kW)

H = amount of peak sunlight (h)



Task C: Calculating the cost of installing solar panels

1) The occupants of a house buy 8 solar panels.
The costs involved with buying and installing the panels are shown below.

- Price per panel: £200
- Storage battery: £4000
- Other required electrical equipment: £1700
- Labour costs: £1500

a) How much will it cost in total to buy and install the solar panels?

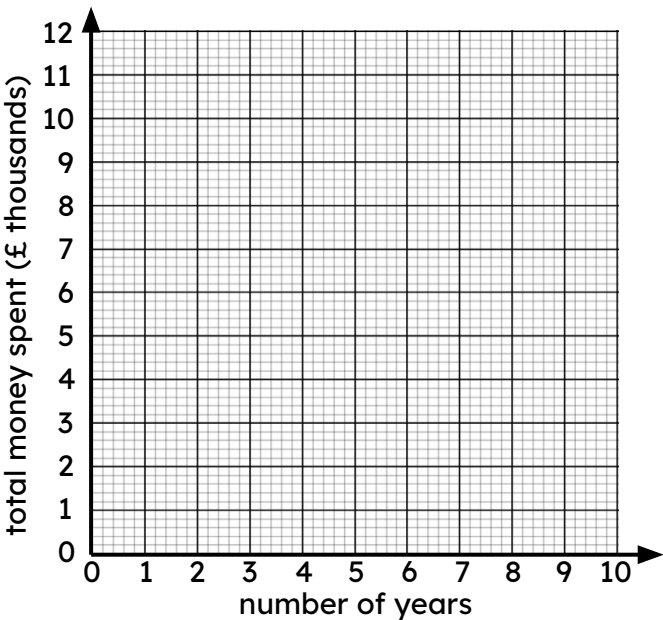
b) Their annual electricity bill without solar panels was £1275.
Their annual electricity bill with solar panels is £350.
How much money do they save per year by having solar panels?

c) How many years will it take for their annual savings to sum to more than the upfront cost of buying and installing the solar panels?

2) The occupants of a house are considering investing in solar panels.
Previously, they obtained all their electricity from their energy supplier company, with an annual bill of £1400.
The solar panels cost £8400 to buy and install.
Once the panels are installed, their annual electricity bill is estimated to be £300.

a) Complete the table below to show the total money spent after each year.

number of years	0	1	2	3	4	5	6	7	8	9	10
total cost without solar (£ thousands)	0	1.4									
total cost with solar (£ thousands)	8.4	8.7									



- b) Use your answer from part (a) to plot a graph showing total money spent after each year **without** solar panels.
- c) Use your answer from part (a) to plot a graph showing total money spent after each year **with** solar panels.
- d) How many years will it take for the solar panels to pay for themselves?